One-level (bond) dissipation on particles: one and two dimensions, at rest and in motion

D. Teaney[1]

[1]*Department of Physics and Astronomy, Stony Brook University, Stony Brook, New York 11794, USA*

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Abstract

We construct a particle discretization of the viscous dissipative function of relativistic hydrodynamics that squares differences of $\mathsf{b} = u/T$ across neighbouring pairs (bonds), rather than squaring a smoothed gradient. In one dimension at rest the bond form is the quadratic form of a graph Laplacian. It is exact for linear profiles on any arrangement, conserves momentum bond by bond, and, unlike the two-level form, has no null mode. In two dimensions the plain bond form dissipates rigid rotation and has a single viscosity. Subtracting a mean-field rotation and dilation from each bond, with the mean field the least-squares fit to the bond differences, fixes both. The rotation is variational, so the faces of each particle carry no net torque and the rotation exerts no mean-field force; the dilation is not, and carries the bulk viscosity. For a moving fluid the same construction holds with three substitutions: bonds are measured by their comoving length, contracted with the comoving metric $\ell^{ij} = \delta^{ij} - v^i v^j$, and the scalar part is removed along the density-frame projector P_{ij}. The continuum limit is then the full density-frame tensor $\kappa^{ijmn}(v)$.

a. Purpose. These notes explain the equations of the construction: the bond form in 1D, its extension to 2D with mean-field rotation and dilation, and to a moving fluid. Numerical tests, Fourier analyses, and comparisons with the code are not included.

I. SETTING

a. Particles and smoothed fields. N particles sit on a periodic line at positions x_a. Each carries an entropy s_a and a momentum m_a. The smoothed entropy and momentum densities are

$$S(x) = \sum_b s_b\, W(x - x_b), \qquad M(x) = \sum_b m_b\, W(x - x_b), \tag{1}$$

with the one-dimensional Wendland C2 kernel

$$W(r) = \frac{3}{4h}\Big(1 - \frac{q}{2}\Big)^4 (1 + 2q), \qquad q = \frac{|r|}{h} < 2, \tag{2}$$

which is normalized, $\int W\, dr = 1$, and has support $|r| < 2h$.

b. Particle volume. Each particle is assigned the volume (in 1D, the length) that its entropy occupies at the local entropy density,

$$V_a = \frac{s_a}{S_a}, \qquad S_a \equiv S(x_a) = \sum_b s_b\, W(x_a - x_b). \tag{3}$$

V_a is the quadrature weight of SPH:

$$\sum_a V_a\, f(x_a) \simeq \int dx\, f(x). \tag{4}$$

On a uniform lattice of spacing a with equal s_a, $V_a = a$ (up to the discrete kernel normalization $\sum_r aW(r) \simeq 1$).

c. Energy, velocity and temperature. The energy H is a function of the smoothed fields. Its derivatives define the particle velocity and the (effective) temperature,

$$\dot{x}_a = \frac{\partial H}{\partial m_a}, \qquad T_a = \frac{\partial H}{\partial s_a}, \qquad \mathsf{b}_a \equiv \frac{\dot{x}_a}{T_a}. \tag{5}$$

In the code, to linear order at rest,

$$\dot{x}_a = \sum_b V_b \, v_b \, W(x_a - x_b), \qquad v_b = \frac{M_b}{w_b}, \tag{6}$$

so b_a is the fluid velocity smoothed *twice*.

d. The dissipative step. Dissipation is described by a dissipative function $K[\mathsf{b}] \geq 0$, quadratic in b. The viscous update at fixed positions is

$$\dot{m}_a = -\frac{\partial K}{\partial \mathsf{b}_a}, \qquad \dot{s}_a = -\mathsf{b}_a \, \dot{m}_a. \tag{7}$$

The second equation keeps the energy fixed:

$$\dot{H} = \sum_a \left(\dot{x}_a \, \dot{m}_a + T_a \, \dot{s}_a \right) = \sum_a \left(\dot{x}_a - T_a \mathsf{b}_a \right) \dot{m}_a = 0, \tag{8}$$

since $T_a \mathsf{b}_a = \dot{x}_a$. The total entropy grows,

$$\dot{S}_{\rm tot} = \sum_a \mathsf{b}_a \frac{\partial K}{\partial \mathsf{b}_a} = 2K \geq 0, \tag{9}$$

using that K is quadratic. Any K built from differences of b conserves total momentum. Such a K is unchanged by $\mathsf{b}_a \to \mathsf{b}_a + c$, so

$$\sum_a \dot{m}_a = -\sum_a \frac{\partial K}{\partial \mathsf{b}_a} = 0. \tag{10}$$

e. The continuum target. In one dimension the strain has a single component $\partial_x \mathsf{b}$, and

$$K = \frac{1}{2} \int dx \; T \kappa \, (\partial_x \mathsf{b})^2, \qquad \kappa \equiv \kappa^{xxxx} = 2\eta \, \frac{d-1}{d} + \zeta \, \Big(= \tfrac{4}{3}\eta + \zeta \text{ for } d = 3 \Big), \tag{11}$$

where d is the physical dimension (the simulation is one-dimensional, but η and ζ are those of a d-dimensional fluid). At rest T is uniform and $T\mathsf{b} = v$, so

$$\dot{M} = \partial_x (\kappa \, \partial_x v). \tag{12}$$

A longitudinal momentum wave $m \propto \cos kx$ therefore decays at

$$\Gamma_{\rm phys}(k) = \frac{\kappa}{w} \, k^2, \qquad w = e + p. \tag{13}$$

(In the full equations this wave is a sound wave and $\Gamma_{\rm phys}$ is twice its amplitude damping rate; here we freeze the particles and study only the dissipative step, so there is no oscillation.)

II. THE TWO-LEVEL FORM

The two-level form discretizes eq. (11) literally. The first level is the standard SPH gradient of b at each particle:

$$\varepsilon_a = \frac{1}{S_a} \sum_b s_b \, (\mathsf{b}_b - \mathsf{b}_a) \, \partial_{x_a} W(x_a - x_b). \tag{14}$$

The second level squares it with the particle volume as the quadrature weight:

$$K_{2L} = \frac{1}{2} \sum_a V_a \, T_a \kappa_a \, \varepsilon_a^2. \tag{15}$$

Differentiating gives the force

$$F_a = \sum_b s_a s_b \, \partial_{x_a} W(x_a - x_b) \, \big[A_a + A_b\big], \qquad A_a = \frac{T_a \kappa_a \, \varepsilon_a}{S_a^2}, \tag{16}$$

which is antisymmetric in each pair, so momentum is conserved.

The force on particle a depends on ε_b at every neighbour b within $2h$. Each ε_b in turn depends on b_c within $2h$ of b. The stencil is therefore $4h$ wide (Fig. 1). More importantly, b enters only through a smoothed gradient. Any pattern of b that this gradient does not see costs nothing in K_{2L} and is not damped.

III. THE ONE-LEVEL (BOND) FORM

a. Bond weight. Define the bond vector and a bond weight

$$r_{ab} = x_b - x_a, \qquad \psi_{ab} = \psi(|r_{ab}|). \tag{17}$$

The one-level form does *not* require ψ to be related to the kernel W. It is an independent function, subject to:

(P1) *Symmetric and non-negative*, $\psi_{ab} = \psi_{ba} \geq 0$. This gives momentum conservation and entropy production bond by bond (below).

(P2) *Connected*: $\psi > 0$ at least for nearest neighbours. Then the only zero mode of K is $\mathsf{b} = \text{const}$, which is the one momentum conservation requires.

(P3) *Short range*, with range ℓ_ψ. The range sets both the long-wavelength error, $O(k^2 \ell_\psi^2)$, and the damping of unresolved modes, $\sim \kappa/(w\ell_\psi^2)$.

(P4) *Continuous*, finite at $r = 0$, and going to zero continuously (preferably with its derivative) at the cutoff. The particles move, and a jump in ψ would make the force jump whenever a neighbour crosses the cutoff.

(P5) *Normalization is irrelevant*: it is divided out by n_a in eq. (19).

In 2D and 3D, ψ must also depend only on $|r|$ (or on the comoving distance when the fluid moves), and its second moment must be isotropic. Ref. [5] enforces this with a moment correction Λ_a. Our default is the Brookshaw weight built from the Wendland kernel,

$$\psi(r) = -\frac{W'(r;h_\psi)}{r} \propto \left(1 - \frac{|r|}{2h_\psi}\right)^3, \qquad h_\psi = h, \tag{18}$$

the weight of the standard SPH second derivative [1]. The range $h_\psi \le h$ is a parameter. A shorter range damps unresolved modes more strongly, but leaves fewer neighbours, which makes the fit noisier and, in 2D, the moment correction of Ref. [5] less stable; the default is $h_\psi = h$.

b. The dissipative function. Instead of squaring a gradient, square the bond differences:

$$\boxed{K_{1L} = \frac{1}{2}\sum_a V_a\, T_a \kappa_a \sum_{b\neq a} w_{ab}\,(\mathsf{b}_b - \mathsf{b}_a)^2, \qquad w_{ab} = \frac{V_b\,\psi_{ab}}{n_a}, \qquad n_a = \sum_{b\neq a} V_b\,\psi_{ab}\,r_{ab}^2.} \tag{19}$$

The normalization n_a is fixed by the continuum limit: the normalized bond weights satisfy

$$\sum_{b\neq a} w_{ab}\, r_{ab}^2 = 1. \tag{20}$$

For a linear profile the bond differences are exact,

$$\mathsf{b}_b - \mathsf{b}_a = \partial_x \mathsf{b}\; r_{ab}, \tag{21}$$

and the inner sum is $(\partial_x \mathsf{b})^2$. Hence

$$K_{1L} = \frac{1}{2}\sum_a V_a\, T_a \kappa_a\, (\partial_x \mathsf{b})^2 \;\longrightarrow\; \frac{1}{2}\int dx\, T\kappa\, (\partial_x \mathsf{b})^2. \tag{22}$$

This holds on *any* particle arrangement, not just a lattice, because n_a is defined by the same sum. For smooth non-linear profiles the next correction to the squared bond difference is odd in r, and cancels on a symmetric neighbourhood. The weights belong to particle a, so $w_{ab} \neq w_{ba}$ in general. K_{1L} is the quadratic form of a graph Laplacian on the bonds, normalized so that for any smooth field ϕ

$$\sum_{b\neq a} w_{ab}(\phi_b - \phi_a) \simeq \frac{1}{2}\,\partial_x^2 \phi. \tag{23}$$

Ref. [5], Sec. VI, collects the general properties of w_{ab}.

c. The force. Write $K_{1L} = \sum_a K_a$, with K_a the bond sum of particle a in eq. (19). For each bond ab define

$$\mathsf{t}_{ab} = -\frac{\partial K_a}{\partial \Delta \mathsf{b}_{ab}} = -V_a T_a \kappa_a\, w_{ab}\, \Delta \mathsf{b}_{ab}, \qquad \Delta \mathsf{b}_{ab} = \mathsf{b}_b - \mathsf{b}_a, \tag{24}$$

The quantity t_{ab} (known as the traction) is the momentum given by a to b along the bond ab.[1] It is the bond analogue of the viscous stress, which is minus the derivative of K with

[1] In much of the literature on elasticity the traction is minus our t_{ab}.

respect to the gradient in the continuum. Along the same bond, b gives t_{ba} to a. The force on a is the net momentum it receives:

$$F_a = -\frac{\partial K_{1L}}{\partial \mathsf{b}_a} = \sum_{b\neq a} f_{ab}, \qquad f_{ab} = -(\mathsf{t}_{ab} - \mathsf{t}_{ba}) = -f_{ba}. \tag{25}$$

Equivalently, each unordered pair appears twice in eq. (19), once from each end, so

$$K_{1L} = \frac{1}{2}\sum_{\text{pairs}} c_{ab}\,(\mathsf{b}_b - \mathsf{b}_a)^2, \qquad c_{ab} = V_a T_a \kappa_a\, w_{ab} + V_b T_b \kappa_b\, w_{ba}, \tag{26}$$

and

$$F_a = \sum_{b\neq a} c_{ab}\,(\mathsf{b}_b - \mathsf{b}_a). \tag{27}$$

This is a sum of pair forces, antisymmetric under $a \leftrightarrow b$. Each pair force pulls b_a and b_b toward each other, i.e. it is diffusion (or "consensus") on the graph of bonds. Each bond separately produces a non-negative entropy, so the second law holds bond by bond. The two-level form has no such local statement.

d. Faces and tractions. For a linear profile, t_{ab} is a momentum flux times an area,

$$\mathsf{t}_{ab} = \Pi_a\, A_{ab}, \qquad \Pi_a = -T_a \kappa_a\, \partial_x \mathsf{b}, \qquad A_{ab} = V_a\, w_{ab}\, r_{ab}. \tag{28}$$

Here Π is the viscous part of the momentum flux, $T^{xx} = p + \Pi$, and A_{ab} plays the role of the area of a face between a and b, with its normal along the bond; $V_a w_{ab}$ is the face area over the bond length. So the traction, the momentum given by a to b, is the flux through the face ab. The definition of n_a is a closure relation for the faces,

$$\sum_{b\neq a} A_{ab}\, r_{ab} = V_a, \tag{29}$$

the discrete form of $\oint x\, d\Sigma = V$. It gives the virial

$$\sum_{b\neq a} \mathsf{t}_{ab}\, r_{ab} = V_a\, \Pi_a, \tag{30}$$

so the stress of a particle is recovered from its faces. For a general profile we take eq. (30) as the definition of Π_a, the viscous stress of particle a. This is the face picture of the grid Metropolis scheme [3], where the momentum crossing a cell wall is the stress times the wall area, and the cell volume is a wall area times a normal length.

IV. TWO DIMENSIONS

We now repeat the construction in two dimensions, again for a fluid at rest. The new feature is that b is a vector and its gradient is a tensor. We first try the direct generalization of the 1D bond form and identify what goes wrong (Sect. IV B). We then fix it by subtracting a mean-field rotation and dilation from each bond (Sect. IV C). To keep the argument simple we assume throughout that every particle's neighbourhood is isotropic, as on a perfect hexagonal lattice. General arrangements are treated in Ref. [5].

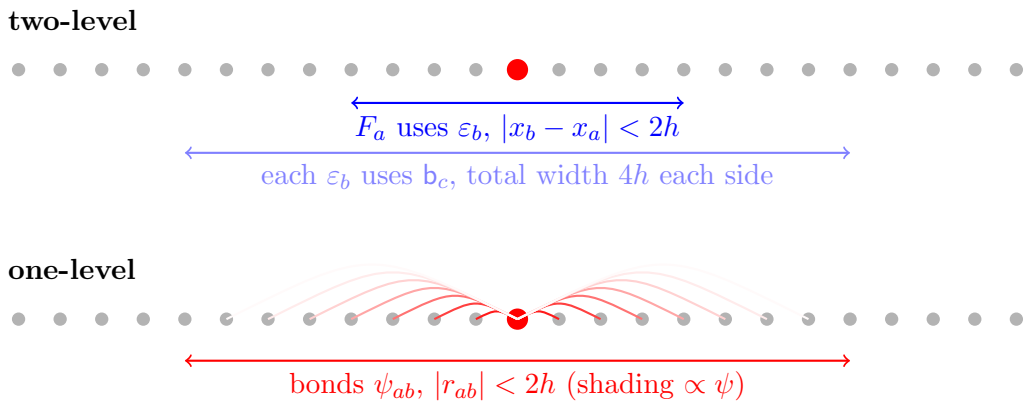


FIG. 1. Stencils for $h/a = 4$. The two-level force on particle a (red) depends on b out to $4h$ through two nested SPH sums. The one-level force is a sum of direct pair forces out to $2h$, weighted by ψ.

A. The continuum target

Write the velocity gradient as the matrix $\nabla\mathsf{b}$ with components

$$\nabla_j \beta_i = \partial_j \beta_i \qquad (\text{row } i,\ \text{column } j), \tag{31}$$

so that along a bond

$$\mathsf{b}(\boldsymbol{x} + \boldsymbol{r}) - \mathsf{b}(\boldsymbol{x}) \simeq (\nabla\mathsf{b})\,\boldsymbol{r}. \tag{32}$$

Split it into its symmetric part S, its antisymmetric part Ω (the local rotation, i.e. the vorticity), and its trace θ (the dilation):

$$S = \tfrac{1}{2}\big(\nabla\mathsf{b} + (\nabla\mathsf{b})^T\big), \qquad \Omega = \tfrac{1}{2}\big(\nabla\mathsf{b} - (\nabla\mathsf{b})^T\big), \qquad \theta = \nabla\cdot\mathsf{b}. \tag{33}$$

We work in d dimensions, the dimension of both the fluid and the simulation; the equations hold in 2D and in 3D. A simulation that resolves fewer directions than the fluid has, such as planar flow of a 3D fluid, is treated in Appendix B. The dissipative function at rest is

$$K = \frac{1}{2}\int d^d x\, T\big[\,2\eta\,\mathring{S}{:}\mathring{S} + \zeta\,\theta^2\big], \qquad \mathring{S} = S - \frac{\theta}{d}\,\mathbb{1}, \tag{34}$$

with $\mathring{S}$ traceless. Two features matter:

- two independent coefficients, the shear viscosity η and the bulk viscosity ζ;
- Ω does not appear: a rigid rotation costs nothing. This is what makes the viscous force conserve angular momentum;

A transverse (shear) wave and a longitudinal wave decay at

$$\Gamma_T = \frac{\eta}{w}k^2, \qquad \Gamma_L = \frac{1}{w}\Big[2\eta\,\frac{d-1}{d} + \zeta\Big]k^2. \tag{35}$$

B. Plain bond form and its two problems

The direct generalization of eq. (19) squares the vector bond differences $\Delta\mathsf{b}_{ab}$:

$$K_{\text{plain}} = \frac{1}{2}\sum_a V_a T_a \eta_a \sum_{b\neq a} w_{ab}\,|\Delta\mathsf{b}_{ab}|^2, \qquad \sum_{b\neq a} w_{ab}\,\boldsymbol{r}_{ab}\boldsymbol{r}_{ab}^T = \mathbb{1}. \tag{36}$$

with w_{ab} defined as in 1D. The second equation is the isotropy assumption; it fixes n_a, the 2D analogue of the 1D bond moment. For a linear field the bond differences are exact, so

$$\sum_b w_{ab}|(\nabla\mathsf{b})\boldsymbol{r}_{ab}|^2 = \nabla_j\beta_i\,\nabla_k\beta_i\sum_b w_{ab} r_{ab}^j r_{ab}^k = \nabla\mathsf{b}{:}\nabla\mathsf{b} = S{:}S + \Omega{:}\Omega, \tag{37}$$

and

$$K_{\text{plain}} \;\to\; \frac{1}{2}\int d^dx\; T\eta\,\big[S{:}S + \Omega{:}\Omega\big]. \tag{38}$$

Compare with eq. (34). There are two problems.

1. *Rotation is dissipated.* Ω appears. A rigid rotation has $S = 0$ but $K_{\text{plain}} \neq 0$. Microscopically, the pair force is along $\Delta\mathsf{b}_{ab}$, not along $\boldsymbol{r}_{ab}$. It exerts a torque, and angular momentum is not conserved.

2. *Only one viscosity.* There is a single coefficient, so the bulk viscosity cannot be set independently of the shear viscosity. A transverse wave happens to come out right: its gradient is half shear, half rotation, and the two errors compensate. But a longitudinal wave ($\Omega = 0$, $S{:}S = k^2$) decays at $\eta k^2/w$. Comparing with eq. (35), the plain form has the fixed bulk viscosity

$$\zeta_{\text{plain}} = -\frac{d-2}{d}\,\eta, \tag{39}$$

which is zero in 2D.

Keeping only the component of the bond difference along the bond, the central form of DPD, cures the first problem but not the second. That form ties ζ to η (the Cauchy relation). With bond differences alone, both problems cannot be fixed at once.

C. Mean-field form: subtract the local rotation and dilation

The cure is simple: before squaring a bond difference, subtract the part of it due to the local rotation and dilation at particle a.

a. The mean field. Define the gradient at particle a as the linear field that best fits its own bond differences, with the same weights as the bonds:

$$(\nabla\mathsf{b})_a \equiv \arg\min_G \sum_{b\neq a} V_b\,\psi_{ab}\,\big|\Delta\mathsf{b}_{ab} - G\,\boldsymbol{r}_{ab}\big|^2. \tag{40}$$

Setting the derivative with respect to G to zero gives

$$(\nabla\mathsf{b})_a \sum_{b\neq a} V_b\,\psi_{ab}\,\boldsymbol{r}_{ab}\boldsymbol{r}_{ab}^T = \sum_{b\neq a} V_b\,\psi_{ab}\,\Delta\mathsf{b}_{ab}\,\boldsymbol{r}_{ab}^T. \tag{41}$$

With the isotropy assumption the bond moment on the left is $n_a \mathbb{1}$, so

$$(\nabla \mathsf{b})_a = \sum_{b \neq a} w_{ab}\, \Delta \mathsf{b}_{ab}\, \boldsymbol{r}_{ab}^T. \tag{42}$$

Split it as in eq. (33):

$$\Omega_a = \tfrac{1}{2}\big((\nabla \mathsf{b})_a - (\nabla \mathsf{b})_a^T\big), \qquad \theta_a = (\nabla \cdot \mathsf{b})_a. \tag{43}$$

For a linear field every bond is fitted exactly, and the fit is the true gradient. With $\psi = -W'/r$, eq. (42) is the usual SPH gradient. The minimization form eq. (40), however, makes sense for any ψ. We call Ω_a and θ_a mean-field quantities: each is an average over the neighbourhood of a.

b. Subtract. Replace $\Delta \mathsf{b}_{ab}$ by

$$\mathcal{D}_{ab}\mathsf{b} = \Delta \mathsf{b}_{ab} - \Big(\Omega_a + \frac{\theta_a}{d}\mathbb{1}\Big)\boldsymbol{r}_{ab}, \tag{44}$$

the *covariant bond difference*. It vanishes when b is, around a, a rigid rotation plus a dilation, i.e. a conformal Killing field: Ω_a and θ_a act as local rotation and Weyl connections (Appendix A). Put the bulk viscosity back in an explicit term:

$$\boxed{K = \sum_a V_a T_a \eta_a \sum_{b \neq a} w_{ab}\, |\mathcal{D}_{ab}\mathsf{b}|^2 + \frac{1}{2}\sum_a V_a T_a \zeta_a\, \theta_a^2.} \tag{45}$$

c. Why it works.

- *Continuum limit.* For a linear field,

$$\mathcal{D}_{ab}\mathsf{b} = \Big(\nabla \mathsf{b} - \Omega - \frac{\theta}{d}\mathbb{1}\Big)\boldsymbol{r}_{ab} = \Big(S - \frac{\theta}{d}\mathbb{1}\Big)\boldsymbol{r}_{ab}, \tag{46}$$

  and the bond sum gives

$$\sum_{b \neq a} w_{ab}\, |\mathcal{D}_{ab}\mathsf{b}|^2 = \operatorname{tr}\Big[\Big(S - \frac{\theta}{d}\mathbb{1}\Big)^2\Big]. \tag{47}$$

  This is $\mathring{S}{:}\mathring{S}$, and with the ζ term eq. (45) reduces exactly to eq. (34). Both viscosities are now correct and independent.

- *Rotation.* For a rigid rotation the fit is Ω_a and $\theta_a = 0$, so every covariant bond difference vanishes and $K = 0$. Angular momentum is conserved.

- *Unresolved modes.* For a mode on the scale of the lattice, the fit eq. (42) sees almost nothing. The subtraction switches off, and K reduces to the plain bond form. That form has no null modes (it is positive for every non-constant b), so short wavelengths are still damped.

d. Resolved plus residual. The argument above used linear fields. For a general field, split each bond difference into the part that the fit explains and the rest,

$$\Delta \mathsf{b}_{ab} = (\nabla \mathsf{b})_a \, \boldsymbol{r}_{ab} + \boldsymbol{\delta}_{ab}. \tag{48}$$

The residual $\boldsymbol{\delta}_{ab}$ is what the fit misses. It vanishes for a linear field, is of order the curvature of b across the star for a smooth field, and is nearly all of $\Delta \mathsf{b}_{ab}$ for an unresolved mode. Because $(\nabla \mathsf{b})_a$ is a least-squares fit, the residuals are orthogonal to the bond vectors; this is the condition that defines the minimum in eq. (40):

$$\sum_{b \neq a} w_{ab} \, \boldsymbol{\delta}_{ab} \, \boldsymbol{r}_{ab}^{T} = 0. \tag{49}$$

The subtraction in eq. (44) is linear in $\boldsymbol{r}_{ab}$, so

$$\mathcal{D}_{ab}\mathsf{b} = \Big(S_a - \frac{\theta_a}{d} \mathbb{1} \Big) \boldsymbol{r}_{ab} + \boldsymbol{\delta}_{ab}, \tag{50}$$

with S_a the symmetric part of the fit. When this is squared and summed with w_{ab}, the cross term is proportional to eq. (49) and vanishes, and the isotropy condition of eq. (36) turns the first term into a trace. K splits exactly into two non-negative pieces:

$$K = \underbrace{\frac{1}{2} \sum_a V_a T_a \left[2\eta \, \mathring{S}_a : \mathring{S}_a + \zeta \, \theta_a^2 \right]}_{\text{resolved}} + \underbrace{\sum_a V_a T_a \eta_a \sum_{b \neq a} w_{ab} \, |\boldsymbol{\delta}_{ab}|^2}_{\text{residual}}. \tag{51}$$

- The *resolved* term is the continuum dissipative function eq. (34) evaluated on the fitted gradient. It is a two-level form, with the fit in place of the SPH gradient.
- The *residual* term is what the bonds add. It is zero for linear fields and small for smooth ones, so it does not change the continuum limit. It is what damps the modes the fit cannot see.

The split is an identity, valid for any field; it only requires that the fit be computed with the same weights as the bonds. Ref. [5] proves it on general arrangements.

In 1D none of this is needed: there is no rotation and a single coefficient κ, so the plain form eq. (19) is already right.

D. Viscous forces

The viscous force on particle a is minus the derivative of the dissipative function, as in eq. (7):

$$\boldsymbol{F}_a = -\frac{\partial K}{\partial \mathsf{b}_a}. \tag{52}$$

K depends on b in two ways. It depends explicitly on the bond differences $\Delta \mathsf{b}_{ab}$ in $\mathcal{D}_{ab}\mathsf{b}$. And it depends implicitly on the neighbours' b through the mean fields Ω_a and θ_a, which are computed from the fit. The force therefore has two parts,

$$\boldsymbol{F}_a = \boldsymbol{F}_a^{\text{bond}} + \boldsymbol{F}_a^{\text{MF}}. \tag{53}$$

The bond force is the derivative at fixed mean fields; the mean-field force is the chain rule through the mean fields. We take them in turn. On a general arrangement the weights are the corrected weights of Ref. [5], so that the isotropy condition of eq. (36) holds exactly; everything below then holds on any arrangement.

a. Bond forces. Write $K = \sum_a K_a$, with K_a the contribution of particle a. As in 1D, eq. (24), define for each bond the traction

$$\mathbf{t}_{ab} = -\frac{\partial K_a}{\partial \Delta \mathsf{b}_{ab}}\bigg|_{\mathrm{MF}} = -2V_a T_a \eta_a \, w_{ab} \, \mathcal{D}_{ab} \mathsf{b}, \tag{54}$$

the momentum given by a to b along the bond ab. It is now a vector, and need not point along the bond. Along the same bond b gives $\mathbf{t}_{ba}$ to a, so the bond force is a sum of antisymmetric pair forces, as in eq. (25):

$$\boldsymbol{F}_a^{\mathrm{bond}} = \sum_{b\neq a} \boldsymbol{f}_{ab}, \qquad \boldsymbol{f}_{ab} = -(\mathbf{t}_{ab} - \mathbf{t}_{ba}) = -\boldsymbol{f}_{ba}. \tag{55}$$

For a linear field the traction is a momentum flux times a face area, as in eq. (28):

$$\mathbf{t}_{ab} = \Pi_a \, \boldsymbol{A}_{ab}, \qquad \Pi_a = -2T_a\eta_a\Big(S_a - \frac{\theta_a}{d}\mathbb{1}\Big), \qquad \boldsymbol{A}_{ab} = V_a \, w_{ab} \, \boldsymbol{r}_{ab}. \tag{56}$$

The faces close,

$$\sum_{b\neq a} \boldsymbol{A}_{ab} \boldsymbol{r}_{ab}^T = V_a \mathbb{1}. \tag{57}$$

As in eq. (30), for a general field we define the stress of particle a by the virial of its faces,

$$V_a \Pi_a = \sum_{b\neq a} \mathbf{t}_{ab} \, \boldsymbol{r}_{ab}^T, \tag{58}$$

which for a linear field is the expression above.

b. The rotation exerts no mean-field force. Define the rotation *variationally*, as the local rigid rotation that minimizes the particle's dissipation:

$$\Omega_a = \arg \min_{\Omega' \in \mathrm{antisym}} \sum_{b\neq a} w_{ab} \Big| \Delta \mathsf{b}_{ab} - \Big(\Omega' + \frac{\theta_a}{d}\mathbb{1}\Big)\boldsymbol{r}_{ab}\Big|^2. \tag{59}$$

The minimizer is exactly the antisymmetric part of the fit (Appendix A), so eq. (43) is unchanged; it has merely been given a variational meaning. The payoff is in the chain rule. The force through the rotation is

$$-\sum_b \frac{\partial K}{\partial \Omega_b} : \frac{\partial \Omega_b}{\partial \mathsf{b}_a} = 0, \tag{60}$$

because at the minimum the derivative of K with respect to each Ω_b vanishes. (This is the envelope theorem: the total derivative of K equals the partial derivative at fixed Ω.) The rotation therefore produces *no* mean-field force. The stationarity condition itself says that the faces of each particle carry no net torque, i.e. that the stress of each particle is symmetric:

$$\sum_{b\neq a} \boldsymbol{r}_{ab} \times \mathbf{t}_{ab} = 0, \qquad \Pi_a = \Pi_a^T. \tag{61}$$

c. The dilation force. Only the dilation, which is not variational, carries a mean-field force. Writing $G_a = (\nabla \mathsf{b})_a$ for short, the chain rule through $\theta_a = \operatorname{tr} G_a$ gives

$$\boldsymbol{F}_a^{\mathrm{MF}} = -\sum_b E_b : \frac{\partial G_b}{\partial \mathsf{b}_a}, \qquad E_a = \frac{\partial K}{\partial G_a} = V_a \Big(T_a \zeta_a \, \theta_a + \frac{\operatorname{tr} \Pi_a}{d} \Big) \mathbb{1}. \tag{62}$$

The first term of E_a is from the bulk term of K. The second is from the dilation subtracted in every bond, and involves the stress eq. (58). From eq. (42), the derivative of the fit is

$$\frac{\partial G_{b,ij}}{\partial \beta_{a,k}} = \delta_i{}^k \, w_{ba} \, r_{ba\,j} \quad (b \neq a), \qquad \frac{\partial G_{a,ij}}{\partial \beta_{a,k}} = -\delta_i{}^k \sum_{b \neq a} w_{ab} \, r_{ab\,j}, \tag{63}$$

where $G_{ij} = \nabla_j \beta_i$ has two lower indices, and the index of the bond vector is lowered with the metric, $r_{ab\,j} = \delta_{jl} \, r_{ab}^l$ at rest. Thus each neighbour of a (for instance particle b) exerts a force

$$\boldsymbol{F}_a^{\mathrm{MF}} = \sum_{b \neq a} \boldsymbol{f}_{ab}^{\mathrm{MF}}, \qquad \boldsymbol{f}_{ab}^{\mathrm{MF}} = \big(w_{ab} E_a + w_{ba} E_b \big) \, \boldsymbol{r}_{ab} = -\boldsymbol{f}_{ba}^{\mathrm{MF}}, \tag{64}$$

directed along the bond. It is antisymmetric, so momentum is conserved. (Had Ω_a been defined non-variationally, for instance from the SPH gradient, Π_a would not be symmetric, and an extra torque term, proportional to the antisymmetric part of Π_a, would appear in E_a. The same happens on particles where the Λ correction fails.)

V. A MOVING FLUID IN TWO DIMENSIONS

We now let the background move with a velocity $\boldsymbol{v}$. The construction of Sect. IV C carries over with three substitutions:

- bonds are measured by their length in the rest frame of the particle;
- bond differences are contracted with the comoving metric ℓ instead of δ;
- the scalar part of the strain is removed along the projector P instead of $\mathbb{1}/d$.

The rotation is subtracted as before.

A. Thermodynamic variables

For a moving fluid the derivative of the energy with respect to the particle entropy is not the rest-frame temperature T. At fixed momentum it is T/γ, so

$$\frac{\partial H}{\partial s_a} = \frac{T_a}{\gamma_a}, \qquad \gamma_a = \frac{1}{\sqrt{1 - v_a^2}}, \qquad \mathsf{b}_a = \frac{\dot{\boldsymbol{x}}_a}{\partial H / \partial s_a} = \frac{\gamma_a \boldsymbol{v}_a}{T_a} = \frac{\boldsymbol{u}_a}{T_a}. \tag{65}$$

So b is the spatial part of the thermal vector $\beta^\mu = u^\mu / T$. At rest the two temperatures coincide. From here on T is the rest-frame temperature.

B. The continuum target

In the density frame the viscous stress is

$$\Pi^{ij} = -T\kappa^{ijmn} S_{mn}, \qquad S_{ij} = \partial_{(i}\beta_{j)}, \tag{66}$$

and the dissipative function is

$$K = \frac{1}{2}\int d^d x\, T\kappa^{ijmn} S_{ij} S_{mn} \tag{67}$$

(Ref. [3]). The tensor $\kappa^{ijmn}(v)$ is built from two objects. The first is the *comoving metric*,

$$\ell^{ij} = \delta^{ij} - v^i v^j, \qquad (\ell^{-1})_{ij} = \delta_{ij} + \gamma^2 v_i v_j, \tag{68}$$

the inverse metric of the comoving coordinates. The second is the *scalar projector*,

$$P_{ij} = -c_s^2 u_i u_j + \frac{1}{d}\big(\delta_{ij} + u_i u_j\big), \qquad \langle P\ell\rangle \equiv P_{ij}\ell^{ij} = 1 - c_s^2 v^2, \tag{69}$$

the direction along which the scalar part of the strain is removed. It encodes the elimination of time derivatives with the Euler equations and the change to the density frame. The shear and bulk parts are

$$\kappa_\eta^{ijmn} = 2\eta\Big[\ell^{(im}\ell^{j)n} - \frac{\ell^{ij}(\ell P\ell)^{mn} + (\ell P\ell)^{ij}\ell^{mn}}{\langle P\ell\rangle} + \frac{\langle P\ell P\ell\rangle}{\langle P\ell\rangle^2}\,\ell^{ij}\ell^{mn}\Big], \tag{70}$$

$$\kappa_\zeta^{ijmn} = \zeta\,\frac{\ell^{ij}\ell^{mn}}{\langle P\ell\rangle^2}. \tag{71}$$

a. A comoving norm. Define the comoving dilation, which at rest is $\nabla\cdot\mathsf{b}$, and the strain with its scalar part removed,

$$\theta = \frac{\ell^{ij}S_{ij}}{\langle P\ell\rangle}, \qquad \mathring{S}_{ij} = S_{ij} - \theta P_{ij}, \tag{72}$$

so that $\mathring{S}$ is traceless in the comoving metric. Then

$$\tfrac{1}{2}T\kappa^{ijmn}S_{ij}S_{mn} = T\eta\ \mathrm{tr}\left(\ell\mathring{S}\ell\mathring{S}\right) + \tfrac{1}{2}T\zeta\theta^2, \tag{73}$$

This is the static form eq. (34) with $\delta^{ij} \to \ell^{ij}$ and θ the comoving dilation eq. (72).

C. Bond form

a. Comoving distance. All positions are taken at the same lab time. The length of the bond ab in the rest frame of a is

$$\rho_{ab}^2 = \boldsymbol{r}_{ab}^T\,\ell_a^{-1}\,\boldsymbol{r}_{ab} = r_{ab}^2 + \gamma_a^2(\boldsymbol{v}_a\cdot\boldsymbol{r}_{ab})^2, \tag{74}$$

the projection of the separation $(0, \boldsymbol{r}_{ab})$ orthogonal to u_a. The bond weight is evaluated at it, $\psi_{ab} = \psi(\rho_{ab})$. A moving fluid is Lorentz contracted along $\boldsymbol{v}$; measuring bonds by ρ_{ab} keeps the stencil round in the rest frame, so each particle has the same neighbours at every velocity.

b. Comoving moment. Weights that are round in ρ give a bond moment proportional to ℓ_a, eq. (14) of Ref. [5]. With the corrected weights of Ref. [5], eq. (41) of Ref. [5], this holds exactly:

$$\sum_{b\neq a} w_{ab}\, \boldsymbol{r}_{ab}\boldsymbol{r}_{ab}^T = \ell_a, \qquad w_{ab} = \frac{V_b\, \psi_{ab}}{n_a}. \tag{75}$$

The face vectors then close on the comoving metric,

$$\boldsymbol{A}_{ab} = V_a w_{ab} \boldsymbol{r}_{ab}, \qquad \sum_{b\neq a} \boldsymbol{A}_{ab}\boldsymbol{r}_{ab}^T = V_a \ell_a. \tag{76}$$

c. Mean field. The fit $(\nabla \mathsf{b})_a$ is defined by eq. (40), which involves no metric. Its rotation Ω_a is the antisymmetric part, as at rest, and its scalar is the comoving dilation,

$$\Omega_a = \tfrac{1}{2}\big((\nabla \mathsf{b})_a - (\nabla \mathsf{b})_a^T\big), \qquad \theta_a = \frac{\ell_a^{ij}\,(\nabla_j \beta_i)_a}{\langle P\ell\rangle_a}. \tag{77}$$

d. Subtract and square. The covariant bond difference becomes

$$\mathcal{D}_{ab}\mathsf{b} = \Delta \mathsf{b}_{ab} - \big(\Omega_a + \theta_a P_a\big)\boldsymbol{r}_{ab}, \tag{78}$$

which reduces to eq. (44) at rest. Contracting it with the comoving metric,

$$\boxed{K = \sum_a V_a T_a \eta_a \sum_{b\neq a} w_{ab}\, \mathcal{D}_{ab}\mathsf{b}\cdot \ell_a \cdot \mathcal{D}_{ab}\mathsf{b} + \frac{1}{2}\sum_a V_a T_a \zeta_a\, \theta_a^2,} \tag{79}$$

where

$$\mathcal{D}\mathsf{b}\cdot\ell\cdot\mathcal{D}\mathsf{b} = (\mathcal{D}\beta)_i\, \ell^{ij} (\mathcal{D}\beta)_j. \tag{80}$$

e. Why it works.

- *Continuum limit.* For a linear field the fit is exact, and

$$\mathcal{D}_{ab}\mathsf{b} = \big(\nabla \mathsf{b} - \Omega - \theta P\big)\boldsymbol{r}_{ab} = \mathring{S}\, \boldsymbol{r}_{ab}. \tag{81}$$

  By eq. (75) the bond sum is

$$\sum_{b\neq a} w_{ab}\, \big(\mathring{S}_{ik}\, r_{ab}^k\big)\, \ell^{ij}\, \big(\mathring{S}_{jl}\, r_{ab}^l\big) = \mathrm{tr}\,\big(\ell \mathring{S} \ell \mathring{S}\big), \tag{82}$$

  and with the ζ term eq. (79) reduces exactly to eq. (73).

- *Rotation.* A rigid rotation has $S = 0$, hence $\theta_a = 0$, and every covariant bond difference vanishes.

- *Unresolved modes.* The fit sees almost nothing, the subtraction switches off, and the bond term reduces to a positive plain bond form contracted with ℓ_a.

f. Resolved plus residual. The split eq. (51) carries over. With the residual of eq. (48), the cross terms vanish for any contraction, since the orthogonality eq. (49) does not involve the metric. With eq. (75),

$$K = \underbrace{\sum_a V_a T_a \left[\eta_a \operatorname{tr}\left(\ell_a \mathring{S}_a \ell_a \mathring{S}_a\right) + \tfrac{1}{2}\zeta_a \theta_a^2\right]}_{\text{resolved}} + \underbrace{\sum_a V_a T_a \eta_a \sum_{b\neq a} w_{ab}\, \boldsymbol{\delta}_{ab} \cdot \ell_a \cdot \boldsymbol{\delta}_{ab}}_{\text{residual}}, \tag{83}$$

with $\mathring{S}_a$ built from the fit. The resolved term is the continuum eq. (73) on the fitted gradient; the residual damps what the fit cannot see.

D. Viscous forces

As at rest, the force $\boldsymbol{F}_a = -\partial K/\partial \mathsf{b}_a$ has a bond part, the derivative at fixed mean fields, and a mean-field part, the chain rule through Ω_a and θ_a:

$$\boldsymbol{F}_a = \boldsymbol{F}_a^{\text{bond}} + \boldsymbol{F}_a^{\text{MF}}. \tag{84}$$

a. Bond forces. The traction is defined as in eq. (54). Since $\mathcal{D}_{ab}\mathsf{b}$ is now contracted with ℓ_a, the traction carries an upper index, raised by ℓ_a:

$$\mathsf{t}_{ab}^i = -\frac{\partial K_a}{\partial(\Delta\beta_{ab})_i}\Big|_{\text{MF}} = -2V_a T_a \eta_a\, w_{ab}\, \ell_a^{ij} (\mathcal{D}_{ab}\beta)_j, \tag{85}$$

the momentum given by a to b along the bond ab. The bond force is again a sum of antisymmetric pair forces,

$$\boldsymbol{F}_a^{\text{bond}} = \sum_{b\neq a} \boldsymbol{f}_{ab}, \qquad \boldsymbol{f}_{ab} = -(\mathbf{t}_{ab} - \mathbf{t}_{ba}) = -\boldsymbol{f}_{ba}. \tag{86}$$

Because ρ_{ab} and ℓ_a use the velocity of a, the two ends of a bond see different weights and $\mathbf{t}_{ab} \neq -\mathbf{t}_{ba}$; the pair force is antisymmetric regardless. For a linear field the traction is a momentum flux times a face area,

$$\mathsf{t}_{ab}^i = \Pi_a^{ik} A_{ab\,k}, \qquad A_{ab\,k} = (\ell_a^{-1})_{kl} A_{ab}^l, \qquad \Pi_a = -2T_a \eta_a\, \ell_a \mathring{S} \ell_a, \tag{87}$$

with $A_{ab\,k}$ the face covector of Ref. [5]. This is the shear stress carried by the bonds; the remainder of the viscous stress, which is proportional to ℓ^{ij}, is carried by the mean field. For a general field the stress of the particle is again the virial of its faces,

$$V_a \Pi_a^{ij} = \sum_{b\neq a} \mathsf{t}_{ab}^i\, r_{ab}^j. \tag{88}$$

b. The rotation exerts no mean-field force. As at rest (Appendix A), consider minimizing the bond term over a local rotation Ω'. As in eq. (A1), only the resolved part depends on Ω'. With $M_a = n_a \ell_a$ and the contraction with ℓ_a it is proportional to a Frobenius norm,

$$\operatorname{tr}\left(\ell X \ell X^T\right) = \left|\ell^{1/2} X \ell^{1/2}\right|^2, \qquad X = \nabla \mathsf{b} - \theta P - \Omega'. \tag{89}$$

It is minimized when

$$\ell^{1/2}\,\Omega'\,\ell^{1/2} = \operatorname{antisym}\!\left(\ell^{1/2}\,\nabla \mathsf{b}\,\ell^{1/2}\right), \tag{90}$$

since P is symmetric and drops out. In 2D every antisymmetric matrix is a multiple of a single matrix J, and $BJB = (\det B)\,J$ for symmetric B. Hence the minimizer is

$$\Omega' = \tfrac{1}{2}\left(\nabla \mathsf{b} - (\nabla \mathsf{b})^T\right) = \Omega_a \qquad \text{(2D, any } \boldsymbol{v}\text{)}, \tag{91}$$

the same subtraction as at rest. (In 3D it is instead the ℓ-weighted antisymmetric part.) So the rotation is variational at any velocity. The derivative of K with respect to Ω_a vanishes at the minimum, so the rotation produces no mean-field force, and the stationarity condition says that the faces carry no net torque: Π_a of eq. (88) is symmetric.

c. The dilation force. Only the dilation, which is not variational, carries a mean-field force. Writing $G_a = (\nabla \mathsf{b})_a$, the chain rule goes through

$$\frac{\partial \theta_a}{\partial G_{a,ij}} = \frac{\ell_a^{ij}}{\langle P\ell \rangle_a}, \tag{92}$$

and gives

$$E_a = \frac{\partial K}{\partial G_a} = e_a\,\ell_a, \qquad e_a = \frac{V_a}{\langle P\ell \rangle_a}\left(T_a \zeta_a\,\theta_a + P_{a,ij}\,\Pi_a^{ij}\right). \tag{93}$$

The first term is from the bulk term of K; the second is from the dilation subtracted in every bond. At rest this reduces to the E_a of eq. (62). With the corrected weights the fit carries a factor ℓ_b^{-1},

$$G_b = \sum_{c \neq b} w_{bc}\,\Delta \mathsf{b}_{bc}\,\boldsymbol{r}_{bc}^T\,\ell_b^{-1}, \tag{94}$$

which cancels the ℓ_b in E_b. Thus each neighbour of a (for instance particle b) exerts a central force,

$$\boldsymbol{F}_a^{\mathrm{MF}} = -\sum_b E_b \!:\! \frac{\partial G_b}{\partial \mathsf{b}_a} = \sum_{b \neq a} \boldsymbol{f}_{ab}^{\mathrm{MF}}, \qquad \boldsymbol{f}_{ab}^{\mathrm{MF}} = \left(w_{ab}\,e_a + w_{ba}\,e_b\right)\boldsymbol{r}_{ab} = -\boldsymbol{f}_{ba}^{\mathrm{MF}}. \tag{95}$$

It is directed along the bond and antisymmetric, so momentum is conserved.

VI. IMPLEMENTATION

This section outlines how the viscous force of Sect. V is computed in the code (`dissipative_force!`, which calls `_bond_dissipation!` in `src/celllist.jl`). The fluid at rest is the special case $\boldsymbol{v}_a = 0$. The routine accumulates the gradient

$$\boldsymbol{g}_a = \frac{\partial K}{\partial \mathsf{b}_a} \tag{96}$$

and returns the force $\boldsymbol{F}_a = -\boldsymbol{g}_a$.

a. Organization. There is a single loop over particles. Each particle a builds its own star of neighbours, computes its weights and its mean field, and then distributes its contributions to $\boldsymbol{g}$ over the star. Both the bond force and the mean-field force of particle a reach only the particles of its own star, so the stencil is $2h$, as in 1D. The ordering of the loop does not matter, because every contribution enters $\boldsymbol{g}$ in equal and opposite pairs.

Algorithm: viscous force $\boldsymbol{F}_a = -\partial K/\partial \mathsf{b}_a$

$\boldsymbol{g}_a \leftarrow 0$ for all a
for each particle a **do**
  star: all $b \neq a$ with $|\boldsymbol{r}_{ab}| < 2h$ (minimum image)
  local data: V_a, T_a, η_a, ζ_a, and ℓ_a, ℓ_a^{-1}, P_a, $\langle P\ell\rangle_a$ from $\boldsymbol{v}_a$
  bare weights: ψ_{ab} at the comoving distance eq. (74)
  correction: solve the 4×4 system eq. (41) of Ref. [5] for Λ_a and n_a
  if a corrected weight is negative **then**
    fall back to $\Lambda_a = 0$
  end if
  weights: w_{ab} of eq. (75)
  fit: bond moment M_a and $G_a = (\nabla \mathsf{b})_a$ by eq. (40)
  mean field: Ω_a and θ_a by eq. (77)
  for each b in the star **do** ▷ bond force
    $\mathcal{D}_{ab}\mathsf{b}$ by eq. (78), traction $\mathbf{t}_{ab}$ by eq. (85)
    $\boldsymbol{g}_a \leftarrow \boldsymbol{g}_a + \mathbf{t}_{ab}$, $\boldsymbol{g}_b \leftarrow \boldsymbol{g}_b - \mathbf{t}_{ab}$
    $V_a \Pi_a \leftarrow V_a \Pi_a + \mathbf{t}_{ab}\, \boldsymbol{r}_{ab}^T$ ▷ virial eq. (88)
  end for
  mean-field stress: $E_a = e_a \ell_a$ by eq. (93)
  for each b in the star **do** ▷ mean-field force
    $\boldsymbol{q}_{ab} = V_b \psi_{ab}\, E_a M_a^{-1} \boldsymbol{r}_{ab}$
    $\boldsymbol{g}_b \leftarrow \boldsymbol{g}_b + \boldsymbol{q}_{ab}$, $\boldsymbol{g}_a \leftarrow \boldsymbol{g}_a - \boldsymbol{q}_{ab}$
  end for
end for
$\boldsymbol{F}_a \leftarrow -\boldsymbol{g}_a$ for all a

b. Remarks.

- *Bond force.* Particle a gives $\mathbf{t}_{ab}$ to b, so it is added to the force on b and subtracted from the force on a. The traction of the reverse bond is added when the loop reaches b. Together they give the pair force of eq. (85).

- *Mean-field force.* The vector $\boldsymbol{q}_{ab}$ is the derivative of the dilation term of particle a with respect to b_b, through the fit. It is written with the full M_a^{-1}, so it remains correct when the Λ correction falls back. When the correction succeeds, $M_a = n_a \ell_a$, and $\boldsymbol{q}_{ab}$ reduces to the central force of eq. (95).

- *Torque term.* The code also adds to E_a a term proportional to the antisymmetric part of Π_a. It vanishes whenever the correction succeeds, since the rotation is then variational, and it keeps the force exact on fallback particles.

- *One dimension.* The 1D path has no mean field. It uses a plain bond term with the full $T\kappa$ on the bonds.

- *Unresolved dimensions.* For planar flow of a 3D fluid the code uses $d = 2$ in P and in the subtracted dilation, with ζ_{eff} of Appendix B.

c. Use in the dissipative step. The force updates the momenta at fixed positions, and the entropy follows from eq. (7):

$$\Delta \boldsymbol{m}_a = \Delta t\, \boldsymbol{F}_a, \qquad \Delta s_a = -\mathsf{b}_a \cdot \Delta \boldsymbol{m}_a. \tag{97}$$

The total entropy grows by $2K\,\Delta t$, and the total momentum is unchanged.

VII. SUMMARY AND OUTLOOK

1. In one dimension at rest, the one-level form eq. (19) is the graph Laplacian of b with weights w_{ab}. It is normalized so that linear profiles are exact on any arrangement. Its force is a sum of momentum-conserving pair forces, the differences of the tractions t_{ab}, and each bond produces entropy.

2. The two-level form sees b only through a smoothed gradient, and patterns that gradient misses, such as the zigzag, are not damped. The bond form has $\mathsf{b} = \text{const}$ as its only zero mode.

3. In 2D the plain bond form dissipates rotation and has only one viscosity. Subtracting the mean-field rotation Ω_a and dilation θ_a, taken from the least-squares fit, fixes both. The rotation is variational: the stress of each particle is symmetric and the rotation exerts no mean-field force. The dilation carries the bulk viscosity and a central mean-field force. For unresolved modes the subtraction switches off, so they remain damped.

4. For a moving fluid the same form holds with the comoving distance, the comoving metric ℓ, and the scalar removed along P. Its continuum limit is the density-frame $\kappa^{ijmn}(v)$, and in 2D the subtracted rotation is the same as at rest.

a. Noise. The one-level form has an obvious fluctuating partner. K_{1L} is a sum over bonds, eq. (26), so the fluctuation–dissipation theorem asks for independent noise on each bond,

$$\delta m_a = \sum_{b \neq a} \sqrt{2c_{ab}\,\Delta t}\;\xi_{ab}, \qquad \xi_{ab} = -\xi_{ba}, \quad \langle \xi_{ab}^2 \rangle = 1, \tag{98}$$

i.e. random tractions on the faces, as in dissipative particle dynamics and the smoothed DPD of Ref. [2]. The two-level noise is instead the SPH divergence of a per-particle random stress, and it shares the null mode. In 2D the noise must also respect the no-torque constraint (Appendix A).

Appendix A: Variational form of the mean field

This appendix shows that the subtracted rotation of Sect. IV C is a minimizer of K itself. This has practical consequences for the force and for the noise. We work in 2D with particle a fixed, and drop the label a where it is unambiguous.

1. The rotation is a minimizer of K

Replace the subtraction in eq. (44) by a general linear one, $A\boldsymbol{r}_{ab}$. As in eq. (51), the residual does not depend on A, and the bond sum of particle a splits as

$$\sum_{b\neq a} w_{ab}\left|\Delta\mathsf{b}_{ab} - A\boldsymbol{r}_{ab}\right|^2 = \operatorname{tr}\left[(\nabla\mathsf{b} - A)(\nabla\mathsf{b} - A)^T\right] + \sum_{b\neq a} w_{ab}|\boldsymbol{\delta}_{ab}|^2. \tag{A1}$$

The Frobenius norm splits orthogonally into symmetric and antisymmetric parts. Subtracting a rotation Ω' and the dilation,

$$A = \Omega' + \frac{\theta}{d}\mathbb{1}, \qquad \operatorname{tr}\left[(\nabla\mathsf{b} - A)(\nabla\mathsf{b} - A)^T\right] = \left|S - \frac{\theta}{d}\mathbb{1}\right|^2 + \left|\Omega - \Omega'\right|^2. \tag{A2}$$

The bond term of K, and therefore K itself (the bulk term does not involve Ω'), is minimized by $\Omega' = \Omega_a$:

$$\boxed{\Omega_a = \arg\min_{\Omega'\,\in\,\text{antisym}} \sum_{b\neq a} V_b\psi_{ab}\left|\Delta\mathsf{b}_{ab} - \left(\Omega' + \frac{\theta_a}{d}\mathbb{1}\right)\boldsymbol{r}_{ab}\right|^2.} \tag{A3}$$

Subtracting the fitted rotation is therefore the same as letting each particle choose the local rigid rotation that minimizes its dissipation. In convex language, K is an *infimal convolution* over local rotations. This is the operation dual, under Legendre transform, to adding a constraint; the rotation is the Lagrange multiplier that enforces zero torque on the tractions. Two consequences follow.

a. 1. No net torque. The stationarity condition of eq. (A3) is

$$\Pi_a = \Pi_a^T, \qquad V_a\Pi_a = \sum_{b\neq a} \mathbf{t}_{ab}\,\boldsymbol{r}_{ab}^T, \quad \mathbf{t}_{ab} = -2V_aT_a\eta_a\,w_{ab}\,\mathcal{D}_{ab}\mathsf{b}. \tag{A4}$$

In 2D the antisymmetric part of the virial is the torque of the tractions about a. The faces of each particle carry *no net torque* about it, for any b field, not just a rigid rotation.

b. 2. No rotational mean-field force. The derivative of K with respect to Ω_a vanishes at the minimum, so the total derivative of K with respect to b equals the partial derivative at fixed Ω_a (the envelope theorem). The chain-rule force through Ω_a, which would be proportional to the antisymmetric part of Π_a, vanishes identically. Only the dilation carries a mean-field force:

$$E_a = V_a\left(T_a\zeta_a\,\theta_a + \frac{\operatorname{tr}\Pi_a}{d}\right)\mathbb{1}. \tag{A5}$$

2. Rotation and dilation as connections

For a linear field the covariant bond difference eq. (44) is the traceless strain applied to the bond, i.e. the conformal Killing operator. Its kernel is the rigid rotations and dilations. Subtracting the rotation compares b_b with b_a after a local rotation: Ω_a is a linearized rotation connection at a. Minimizing over it, eq. (A3), integrates the connection out, and its equation of motion is the no-torque condition. The subtracted dilation is likewise a Weyl connection, as in the Weyl-covariant form of conformal hydrodynamics [4]. Weyl rescaling, however, is a symmetry of the shear part only, and the bulk viscosity breaks it. That is why θ_a cannot be integrated out and the bulk term is added explicitly. Unlike lattice gauge fields, Ω_a and θ_a belong to the particle, not to the bond.

3. Dual picture and the noise

The Legendre dual of the dissipative function is the dissipation potential in terms of forces. Under this duality the infimum over local rotations in eq. (A3) becomes a constraint on the tractions: the faces of each particle must carry no net torque. For the stochastic scheme this suggests the form of the noise. Draw an independent random traction $\delta\mathbf{t}_{ab}$ on each face, as in DPD, and project it onto the torque-free subspace around each particle. In the grid scheme [3] the noise is instead a tensor ξ^{ij} at the cell corners, contracted with the wall areas $d\Sigma_j$; on particles the faces are the primary objects, and the noise has one index. The dilation, which is not variational, needs its own explicit mean-field noise of strength set by ζ.

Appendix B: Unresolved dimensions

In the main text d is the dimension of both the fluid and the simulation. Here we separate the two. Let

- d be the dimension of the fluid, which fixes the traceless strain and the projector P;
- $d_s \leq d$ be the dimension of the simulation, the number of directions that are resolved.

Nothing varies along the other $d - d_s$ directions, and the velocity lies in the resolved ones. An example is planar flow of a three-dimensional fluid, $d = 3$ and $d_s = 2$. Write tr_d for a trace over all d directions and tr_{d_s} for a trace over the resolved ones only. The bonds see only the resolved components, so the bond construction produces tr_{d_s}, while the dissipative function of the fluid is a tr_d.

a. The result. Everything in the main text can be used with $d \to d_s$, i.e. as for a d_s-dimensional fluid, provided the bulk viscosity is replaced by

$$\boxed{\zeta \;\to\; \zeta_{\text{eff}} = \zeta + 2\eta\,\varepsilon, \qquad \varepsilon = \frac{1}{d_s} - \frac{1}{d}.} \tag{B1}$$

For $d = 3$ and $d_s = 2$, $\varepsilon = 1/6$ and $\zeta_{\text{eff}} = \zeta + \eta/3$.

b. At rest. The strain S has components only in the resolved directions, with trace θ. The traceless strain of the fluid and of the simulation are

$$\mathring{S}_d = S - \frac{\theta}{d}\mathbb{1}_d, \qquad \mathring{S}_{d_s} = S - \frac{\theta}{d_s}\mathbb{1}_{d_s}. \tag{B2}$$

Their norms are

$$\mathrm{tr}_d\left(\mathring{S}_d\mathring{S}_d\right) = S{:}S - \frac{\theta^2}{d}, \qquad \mathrm{tr}_{d_s}\left(\mathring{S}_{d_s}\mathring{S}_{d_s}\right) = S{:}S - \frac{\theta^2}{d_s}, \tag{B3}$$

so

$$\mathrm{tr}_d\left(\mathring{S}_d\mathring{S}_d\right) = \mathrm{tr}_{d_s}\left(\mathring{S}_{d_s}\mathring{S}_{d_s}\right) + \varepsilon\,\theta^2. \tag{B4}$$

The last term is the shear of the unresolved directions: a dilation in the resolved directions is, for the d-dimensional fluid, also a traceless deformation that involves the others. Multiplying by 2η and adding the bulk term,

$$2\eta\ \mathrm{tr}_d\left(\mathring{S}_d\mathring{S}_d\right) + \zeta\theta^2 = 2\eta\ \mathrm{tr}_{d_s}\left(\mathring{S}_{d_s}\mathring{S}_{d_s}\right) + \zeta_{\mathrm{eff}}\,\theta^2. \tag{B5}$$

The left side is the fluid; the right side is a d_s-dimensional fluid with the same shear viscosity and the bulk viscosity ζ_{eff}.

c. Moving. The dilation eq. (72) does not depend on the dimension, because $\langle P\ell\rangle = 1 - c_s^2 v^2$ for any d. Write P_d and P_{d_s} for the projector built with $1/d$ and $1/d_s$. In the resolved directions they differ by

$$P_d - P_{d_s} = -\varepsilon\left(\delta_{ij} + u_i u_j\right) = -\varepsilon\,(\ell^{-1})_{ij}, \tag{B6}$$

so the two traceless strains differ by

$$\mathring{S}_d = \mathring{S}_{d_s} + \varepsilon\,\theta\,\ell^{-1} \qquad \text{(resolved block)}. \tag{B7}$$

Inserting this into the comoving norm,

$$\mathrm{tr}_{d_s}\left(\ell\mathring{S}_d\ell\mathring{S}_d\right) = \mathrm{tr}_{d_s}\left(\ell\mathring{S}_{d_s}\ell\mathring{S}_{d_s}\right) + 2\varepsilon\,\theta\,\ell^{ij}(\mathring{S}_{d_s})_{ij} + \varepsilon^2\theta^2\ \mathrm{tr}_{d_s}\left(\ell\ell^{-1}\ell\ell^{-1}\right). \tag{B8}$$

The middle term vanishes, since $\mathring{S}_{d_s}$ is traceless in the comoving metric, and the last trace is d_s. In each of the $d - d_s$ unresolved directions $\ell = 1$, $P = 1/d$ and $S = 0$, so the diagonal component of $\mathring{S}_d$ is $-\theta/d$, contributing θ^2/d^2. Altogether

$$\mathrm{tr}_d\left(\ell\mathring{S}_d\ell\mathring{S}_d\right) = \mathrm{tr}_{d_s}\left(\ell\mathring{S}_{d_s}\ell\mathring{S}_{d_s}\right) + \left[\varepsilon^2 d_s + \frac{d-d_s}{d^2}\right]\theta^2, \tag{B9}$$

and the bracket is

$$\varepsilon^2 d_s + \frac{d-d_s}{d^2} = \frac{(d-d_s)^2}{d^2 d_s} + \frac{d-d_s}{d^2} = \frac{d-d_s}{d\,d_s} = \varepsilon. \tag{B10}$$

This is eq. (B4) again, now at any velocity, and eq. (B1) follows as at rest.

d. Beyond linear fields. One could instead keep d in P and in the subtracted dilation, and put the shear of the unresolved directions into the coefficient of θ^2. The two choices give the same K for *any* field, not only a linear one. By eq. (51), K is the continuum dissipative function on the fitted gradient plus the residual. The residual does not depend on what linear part is subtracted, and the resolved parts agree by eq. (B4). This requires the corrected weights; on particles where the correction falls back, the two differ slightly.

e. One dimension. For $d_s = 1$ the traceless strain of the simulation vanishes, so there is no shear term, and

$$\zeta_{\mathrm{eff}} = \zeta + 2\eta\,\frac{d-1}{d} = \kappa, \tag{B11}$$

the coefficient of eq. (11).

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